

COS10005

Web Development

Lecture 5

CSS Part 2

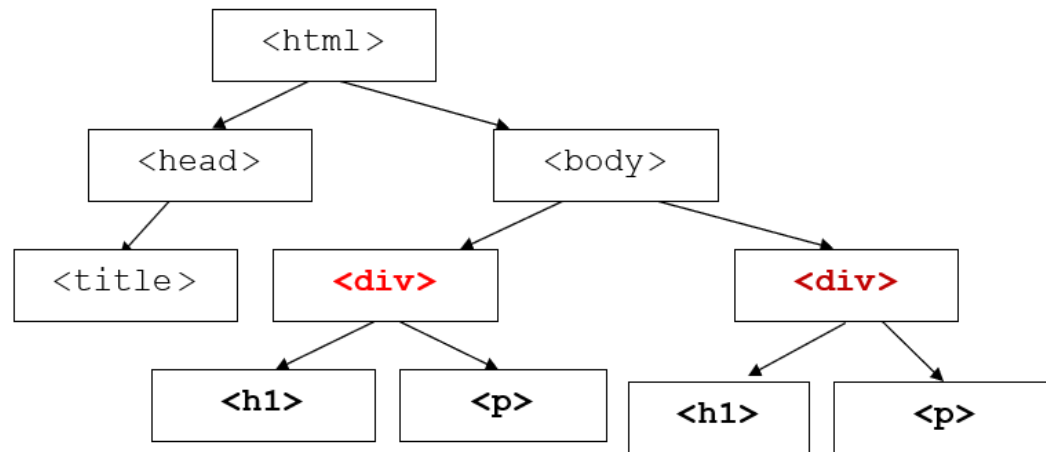
Contents

- HTML elements behavior
 - Blocked
 - Inline
- Ways to combine CSS & HTML
- CSS Web Typography
 - Colour
 - Fonts
 - Text
- CSS Model
 - Visual Formatting
 - Box Model
- CSS Page Layout
 - Fixed and Fluid
 - Nav and Div
- CSS Table

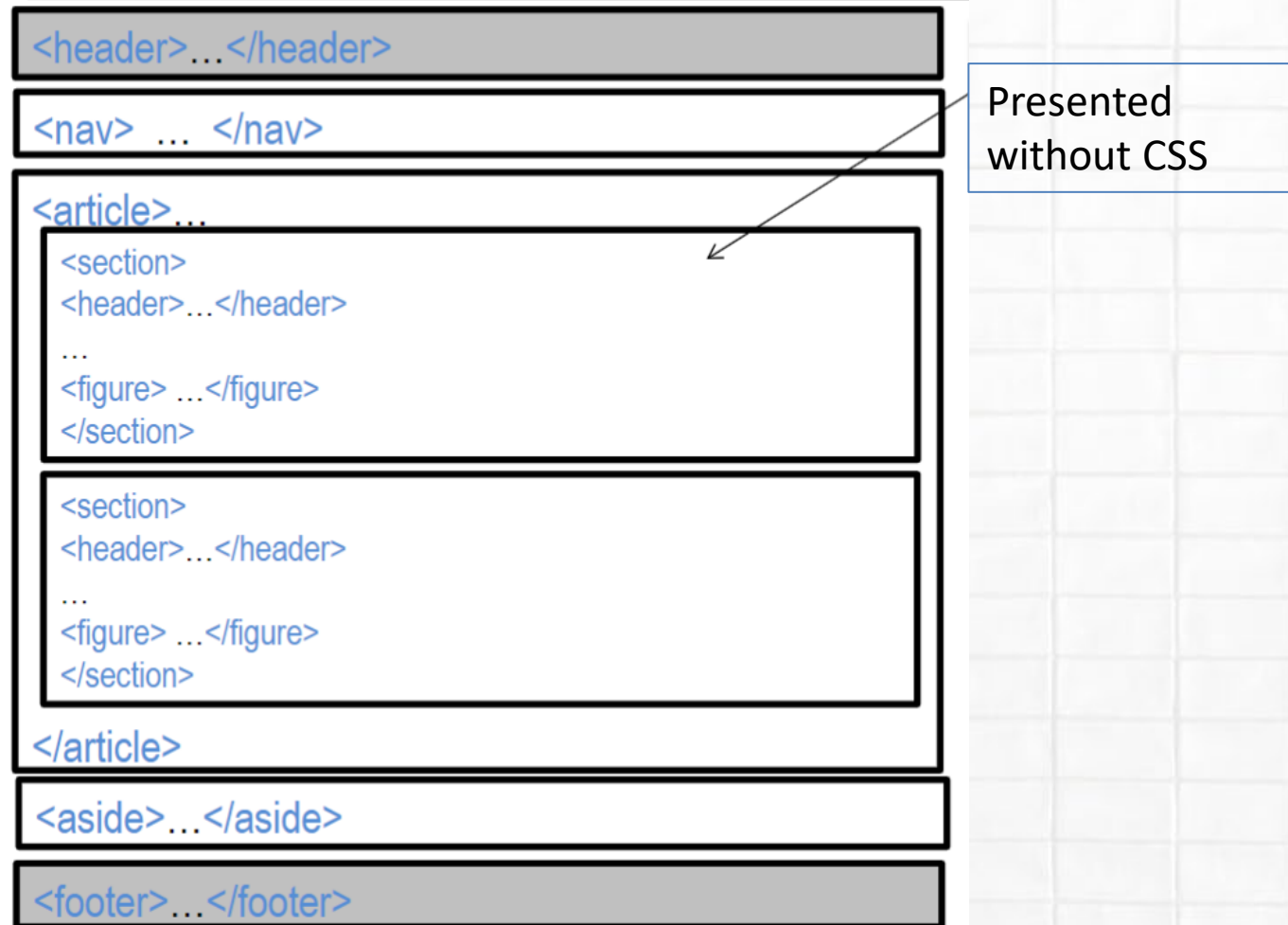
Division

- The **html** element – the ‘**root**’ element of any html document usually contains only two children **head** and **body**.
- The **head** then contains the **title**, and other ‘head’ elements.
- The **body** can contain many other elements, grouped by **div** elements.

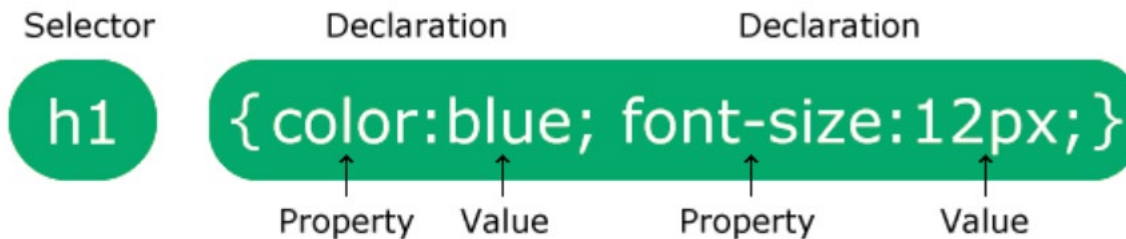
```
<html>  
<head>  
  <title>...</title>  
</head>  
<body>  
  <div>  
    <h1>...</h1>  
    <p>...</p>  
  </div>  
<div>
```



Example of Web Page Structure



CSS Format



- H1 is a selector in CSS (it points to the HTML element you want to style).
- Color is a property, and blue is the property value.
- Font-size is a property and 12px is the property value.

Elements Behavior

- Block elements
 - Create a block-level box in the document flow.
 - They start on a new line.
 - They take up the full available width of their container, and can have padding, margins, and borders applied to them.
 - Some examples of block-level elements include: `<p>`, `<div>`, `<header>`, `<article>`, etc.
- In-line elements

In-line elements

Elements that are formatted on a line-by-line basis, meaning they only take up the space necessary to display their content, and do not start on a new line.

Some examples of inline-level elements include:

- `<a>` (anchor links)
- `<span>` (generic inline container)
- `<img>` (images)
- `<input>` (form input fields)
- `<label>` (form labels)
- `<button>` (buttons)
- `<strong>` (bold text)
- `<em>` (italicized text)

In-line elements

- Inline-level elements can be styled with CSS properties such as color, font-size, and text-align.
- **They cannot be given width or height properties, and margins and padding values only apply to their horizontal axis (left and right).**
- To make an inline-level element behave as a block-level element (taking up the full width of its parent container and starting on a new line), you can use the CSS display property and set it to 'block'.

```
<a href="#" class="my-link">My Link</a>
```

```
.my-link {  
  display: block;  
  width: 200px; /* optional: set a width to make the element take up a specific amount of space */  
  background-color: #ccc; /* optional: add a background color to see the effect */  
}
```

Three Ways to Insert CSS

- There are three (3) ways of inserting a style sheet:
 - External CSS
 - Internal CSS
 - Inline CSS

Example of External CSS file

```
/* This is a comment */

/* Global styles */
body {
  font-family: Arial, sans-serif;
  font-size: 16px;
  color: #333;
}

/* Header styles */
header {
  background-color: #f2f2f2;
  padding: 20px;
}

header h1 {
  font-size: 36px;
  color: #333;
  margin: 0;
}
```

```
/* Navigation styles */
nav {
  background-color: #333;
  color: #fff;
  padding: 10px;
}

nav ul {
  list-style: none;
  margin: 0;
  padding: 0;
}

nav li {
  display: inline-block;
  margin: 0 10px;
}

nav a {
  color: #fff;
  text-decoration: none;
}
```

```
nav a:hover {
  text-decoration: underline;
}

/* Main content styles */
main {
  padding: 20px;
}

section {
  margin-bottom: 20px;
}

section h2 {
  font-size: 24px;
  margin-top: 0;
}

section p {
  line-height: 1.5;
}
```

```
/* Footer styles */
footer {
  background-color: #f2f2f2;
  padding: 20px;
  text-align: center;
  font-size: 12px;
  color: #666;
}
```

Link CSS and HTML

```
<!DOCTYPE html>
<html>
  <head>
    <title>My HTML Page</title>
    <link rel="stylesheet" type="text/css" href="style.css">
  </head>
  <body>
    <header>
      <h1>Welcome to my website</h1>
    </header>
    <nav>
      <ul>
        <li><a href="#">Home</a></li>
        <li><a href="#">About Us</a></li>
        <li><a href="#">Contact</a></li>
      </ul>
    </nav>
    <main>
      <section>
        <h2>Section 1</h2>
        <p>Lorem ipsum dolor sit amet, consectetur adipiscing elit. Fusce eu
      </section>
```

- To style an HTML page, you need to link the CSS file to your HTML document.
- Do this by adding a <link> element in the <head> section of your HTML document.

Example of Internal CSS

```
<!DOCTYPE html>
<html>
  <head>
    <title>Internal CSS Example</title>
    <style>
      /* Internal CSS styles */
      h1 {
        color: red;
        font-size: 32px;
        font-weight: bold;
      }
      p {
        color: blue;
        font-size: 18px;
      }
    </style>
  </head>
  <body>
    <h1>This is a heading</h1>
    <p>This is a paragraph.</p>
  </body>
</html>
```

Example of in-line CSS

```
<!DOCTYPE html>
<html>
  <head>
    <title>Inline CSS Example</title>
  </head>
  <body>
    <h1 style="color: red; font-size: 32px; font-weight: bold;">This is a heading</h1>
    <p style="color: blue; font-size: 18px;">This is a paragraph.</p>
  </body>
</html>
```

Web Topography: Dimension

- Absolute is used for printed media.

Unit	Abbr	Description	Example
centimetre	cm	metric centimetre	p {padding :1cm;}
inch	in	US inch	p {margin: 1.25in;}
millimetre	mm	metric millimetre	p {word-spacing: 10mm;}
pica	pc	Equal to 12 points	p {font-size: 20pc;}
point	pt	Equal to 72 points in an inch	p {font-size: 24pt;}

1 pc = 0.423cm, 12 pc = 5.08cm

1 pt = 0.03528 cm, 12 pt = 0.423 cm

Web Topography: Measurements

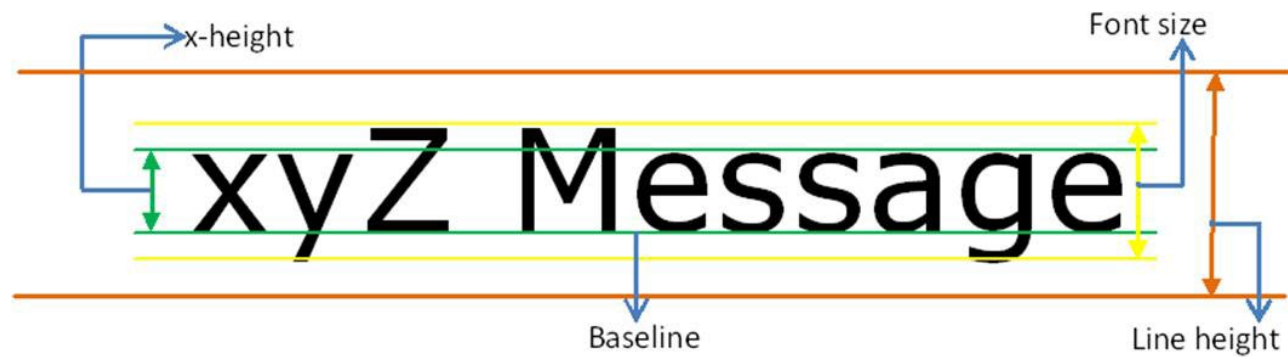
- Relative is used for webpage.

Unit	Abbr	Description	Example
EM	em	Height of the current font size	p {padding :2em;}
Percentage	%	Percentage as relative to parent element	p {line-height: 100%;}
Ex	ex	Height of letter x in the current font	p {margin: 25ex;}
Pixel	px	Pixel size of screen	p {font-size : 12px;}

1 em = 0.42cm, 2 em = 0.84cm

1 ex = 0.2cm, 2 ex = 0.4cm

1 px = 0.0265 cm, 12 px = 0.318 cm



Web Topography: Colour

- Color: <color values>;
- Color values can be in text or numerical format
 - aqua | black | blue | fuchsia | gray | grey | green | lime | maroon | navy | olive | purple | silver | red | teal | white | yellow
 - #<6-digit hexadecimal> representing rrggbb
 - Rgb | rgba | hsl | hsla (<0-255>, <0-255>, <0-255>)
 - Example:
 - h1 {color : blue;}
 - h1 {color : #0000FF;}
 - h1 {color : rgb (0, 0, 255);}

Web Topography: Fonts

- A generic font refers to the font's general appearance such as: "serif", "sans-serif", "monospace", "cursive" or "fantasy".

Five Generic Fonts	Font Samples			
	serif	defg	defg	defg
	sans-serif	defg	defg	defg
	monospace	defg	defg	defg
	cursive	defg	<i>defg</i>	defg
	fantasy	<i>defg</i>	defg	DEFG

FF

The lines on letters are called "serifs".
"sans-serif" means "without serifs".

Web Topography: Fonts

- A specific font is a font such as “Times New Roman”, “Arial”, or “Courier New”. Specifies fonts are installed on a user’s computer, so availability depends on the user’s machine.

Web Topography: Fonts Principles

- Use commonly available fonts
 - Times New Roman - Arial - Courier New
- User font-family safely
 - article {font-family: Times New Roman, serif}
 - article {font-family: Arial, sans-serif}
 - article {font-family: Courier New, monospace}

Times New Roman
Garamond
Georgia

} serif

Arial
Verdana
Trebuchet

} sans-serif

Arial
Verdana
Trebuchet

} serif

Web Topography: Fonts

- font-family: ;
 - font-family: Arial, “Times New Romans”;
- font-size: <value> | <keyword>;
 - H1 {font-size: 0.8em;}
 - H1 {font-size: small;}
 - Xx-small/x-small/small/medium/large/x-large/xx-large
- font-style: normal | italic;
 - H1 {font-style:italic;}

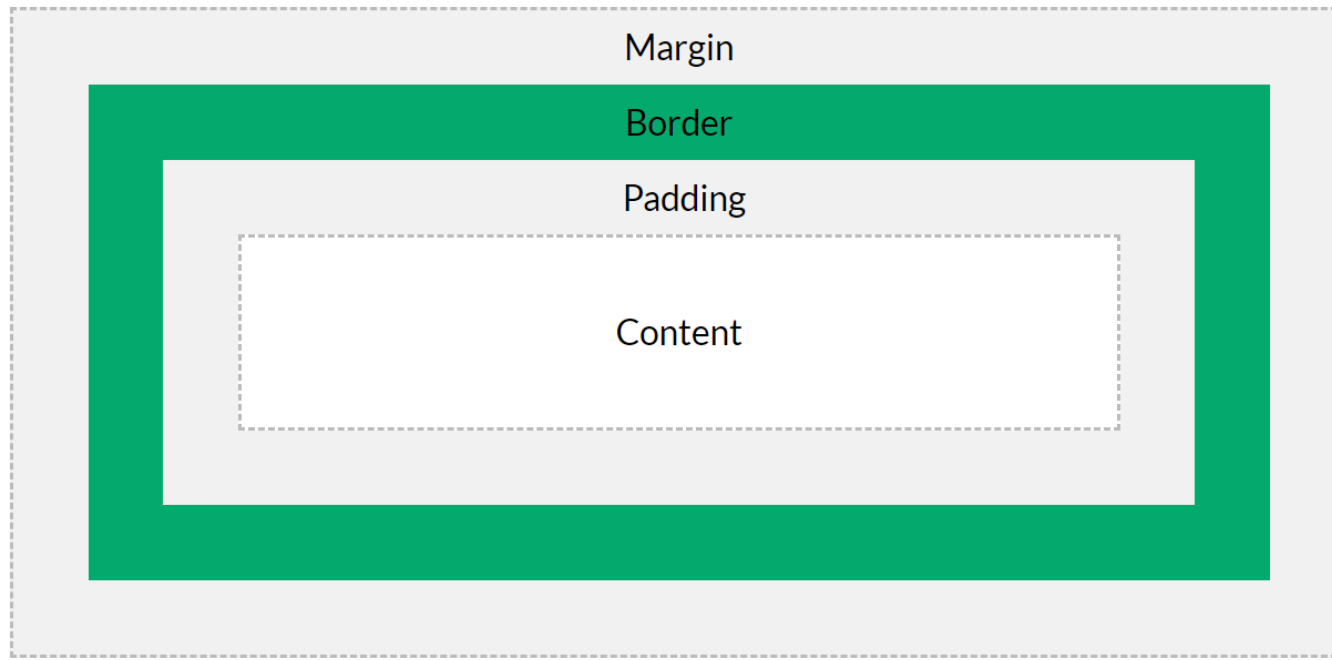
Web Topography: Fonts

- `font-variant: normal | small-caps;`
 - `font-variant: small-caps;`
- `font-weight: normal | bold | bolder | lighter | 100 | 200 | 300 | 400 | 500 | 600 | 700 | 800 | 900;`
 - Font-weight: bold;
- `font: <short cut property>;`
 - Font: bold 1.2em Arial;

font-weight, font-size and font-family on one line.

CSS Box Model

- The CSS box model is essentially a box that wraps around every HTML element. It consists of: margins, borders, padding, and the actual content.



Elements of Box Model

- **Content** – The content of the box, where text and images appear.
- **Padding** – Clears an area around the content. The padding is transparent.
- **Border** – A border that goes around the padding and content.
- **Margin** – Clears an area outside the border. The margin is transparent.

Example of Box Model

```
<!DOCTYPE html>
<html>
<head>
<style>
div {
  background-color: lightgrey;
  width: 300px;
  border: 15px solid green;
  padding: 50px;
  margin: 20px;
}
</style>
</head>
<body>

<h2>Demonstrating the Box Model</h2>

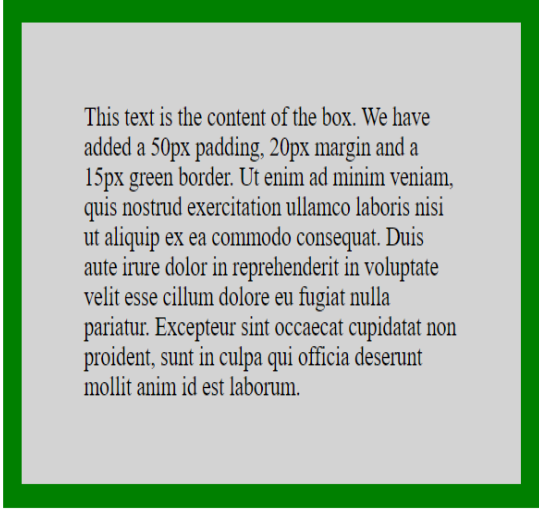
<p>The CSS box model is essentially a box that wraps around every HTML element.
It consists of: borders, padding, margins, and the actual content.</p>

<div>This text is the content of the box. We have added a 50px padding, 20px
margin and a 15px green border. Ut enim ad minim veniam, quis nostrud
exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute
irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat
nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa
qui officia deserunt mollit anim id est laborum.</div>

</body>
</html>
```

Demonstrating the Box Model

The CSS box model is essentially a box that wraps around every HTML element. It consists of: borders, padding, margins, and the actual content.



This text is the content of the box. We have added a 50px padding, 20px margin and a 15px green border. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

CSS Page Layout

- Fix (the size of the webpages is set to specific width or pixels)
 - Fixed width in pixels.
 - Text doesn't scroll down when browser windows in minimized.
 - Independent of screen size.
 - Horizontal scroll will come when screen size is less than width of main container.
 - Often used when designer want to have complete control over the placement of the page elements, such as images, text, and navigation menus.

Fixed layout

```
<html>
<head>
  <title>Fixed Layout Example</title>
  <style>
    .container {
      width: 960px; /* set the container width to 960 pixels */
      margin: 0 auto; /* center the container horizontally */
    }
    .header {
      background-color: #ccc;
      height: 100px;
    }
    .content {
      background-color: #f1f1f1;
      height: 500px;
      padding: 20px;
    }
    .footer {
      background-color: #ccc;
      height: 50px;
    }
  </style>
</head>
<body>
  <div class="container">
    <header class="header">
      <h1>Fixed Layout Example</h1>
    </header>
    <main class="content">
      <p>This is an example of a fixed layout. The container has a fixed width
    </main>
    <footer class="footer">
```

CSS Page Layout

- Fluid (adjust webpage based on the size of the browser window)
 - Width in percentage (%).
 - Text scroll down when zoom in or zoom out.
 - Dependent of screen size.
 - Horizontal scroll will never come on any screen unless any fixed content is inside.

Fluid Layout

```
<html>
<head>
  <title>Fluid Layout Example</title>
  <style>
    .container {
      max-width: 100%; /* set the container width to 100% of the viewport */
      margin: 0 auto; /* center the container horizontally */
      padding: 20px;
      box-sizing: border-box; /* include padding in the element's total width
    }
    .header {
      background-color: #ccc;
      height: 100px;
    }
    .content {
      background-color: #f1f1f1;
      min-height: 500px;
      padding: 20px;
    }
    .footer {
      background-color: #ccc;
      height: 50px;
    }
  </style>
</head>
<body>
  <div class="container">
    <header class="header">
      <h1>Fluid Layout Example</h1>
    </header>
    <main class="content">
      <p>This is an example of a fluid layout. The container width is set to 1
    </main>
```

Difference between Nav and Div

- Nav has a semantic meaning browsers, assistive technology, bot that will read this tag will know that the role of this element is a navigation and can act accordingly.
- DIV doesn't carry any semantic, it is only used to group elements for layout purpose to create some containers.
- Note: Web developers **use <div>** elements to group together **HTML** elements and apply CSS styles to many elements at once.
- Note: Google crawl uses semantic to identify elements of our websites (use NAV to improve your ranking).
- Note: you can embed a <nav> inside <div>.

CSS Tables - Table

```
table {  
  width: 100%;  
}  
  
th {  
  height: 70px;  
  color: white;  
}
```

Creates a table with the width of 100%, and the height of the <th> elements to 70px

```
th, td {  
  padding: 15px;  
  text-align: left;  
  background-color: black;  
  color: white;  
}
```

Padding controls the space between content and boarder

Firstname	Lastname	Savings
Peter	Griffin	\$100
Lois	Griffin	\$150
Joe	Swanson	\$300

CSS Tables – Responsive Table

- A responsive table will display a horizontal scroll bar if the screen too small to display the full content.
- Add a container element (like <div>) with **overflow-x:auto** around the <table> element to make it responsive:

```
<div style="overflow-x:auto;">
  <table>
    <!-- table content -->
  </table>
</div>
```

Example of Responsive Table

```
<html>
<head>
<style>
table {
  border-collapse: collapse;
  width: 100%;
}

th, td {
  text-align: left;
  padding: 8px;
}

tr:nth-child(even) {background-color: #f2f2f2;}
</style>
</head>
<body>

<h2>Responsive Table</h2>
<p>A responsive table will display a horizontal scroll bar if the screen is too
small to display the full content. Resize the browser window to see the effect:</p>
<p>To create a responsive table, add a container element (like div) with <strong>overflow-
x:auto</strong> around the table element:</p>

<div style="overflow-x: auto;">
  <table>
    <tr>
      <th>First Name</th>
      <th>Last Name</th>
      <th>Points</th>
      <th>Points</th>
      <th>Points</th>
      <th>Points</th>
      <th>Points</th>
      <th>Points</th>
      <th>Points</th>
      <th>Points</th>
      <th>Points</th>
    </tr>
```

Responsive Table

A responsive table will display a horizontal scroll bar if the screen is too small to display the full content. Resize the browser window to see the effect:

To create a responsive table, add a container element (like div) with **overflow-x:auto** around the table element:

First Name	Last Name	Points	Points	Points	Points	Points	Points	Points	Points	Points	Points
Jill	Smith	50	50	50	50	50	50	50	50	50	50
Eve	Jackson	94	94	94	94	94	94	94	94	94	94
Adam	Johnson	67	67	67	67	67	67	67	67	67	67

Wake brain

- Which of the following is not a valid CSS selector?



#id-selector



.class-selector



!important-selector



element-selector

Wake brain

- What is the purpose of the CSS box model?



To define the layout of a webpage



To define the size and shape of a webpage



To define the dimensions of a webpage



To define the border and padding of an element

Wake brain

- Which of the following is the correct way to apply inline CSS to an HTML element?



`<p class="my-paragraph" style="font-size: 18px;">Lorem ipsum dolor sit amet</p>`



`<p class="my-paragraph" href="style.css">Lorem ipsum dolor sit amet</p>`



`<p class="my-paragraph" rel="stylesheet" href="style.css">Lorem ipsum dolor sit amet</p>`



`<p class="my-paragraph">Lorem ipsum dolor sit amet</p>`

Wake brain

- What is the correct syntax for using an external CSS file in HTML?



`<style src="style.css">`



`<link rel="stylesheet" href="style.css">`



`<script src="style.css">`



`<style href="style.css">`

Thank You